Lecture 11: From Two To Multiples Treatment

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Outline

01

ANALYSIS OF VARIANCE

From two treatments to several: one F-test for all groups, then pairwise comparisons

02

ANOVA AS A REGRESSION

Coding a category as dummy variables — and getting the same F from the Regression tool

03

ONE CONTINUOUS PREDICTOR

Floor area enters; the ANOVA block of a regression output; $F = t^2$

04

MULTIPLE REGRESSION

Both kinds of variable together: partial coefficients, t versus F, adjusted R^2

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ANALYSIS OF VARIANCE

From Two Treatments to Several

The chain now tests three layouts. Why not simply run a t-test for every pair?

14%

3 GROUPS · 3 PAIRS

Chance of at least one false alarm when every group is identical, each test at $\alpha = 0.05$: $1 - 0.95^3$.

26%

4 GROUPS · 6 PAIRS

$1 - 0.95^6$. One false alarm in four sets of comparisons.

76%

8 GROUPS · 28 PAIRS

$1 - 0.95^{28}$. A 'significant' pair is now almost guaranteed.

$\alpha = 0.05$ holds for one test. Across many tests, the chance of some false alarm grows fast.

One Question for All k Groups

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$ versus H_1 : not all μ_i are equal. One test, so the false-alarm rate is α — exactly.

The alternative is not that every mean differs from every other. It is only that at least one differs. Analysis of variance (ANOVA) answers this single question; a significant result says that the treatments are not all alike, and says nothing yet about which of them differ.

Example: Three Store Layouts

Each layout is used in four stores · weekly sales, million VND

Layout	Weekly sales of the four stores	Mean $\bar{x}_i$	Sample variance s_i^2
A	48 · 52 · 50 · 46	49	6.67
B	55 · 57 · 53 · 59	56	6.67
C	50 · 54 · 52 · 48	51	6.67
All 12 stores	grand mean $\bar{x} = 52$		

The three means differ. Is the difference larger than the spread inside each layout would produce by chance?

Two Sources of Variation

MECHANISM

Between groups and within groups

Between groups: how far the group means sit from the grand mean. Within groups: how far observations sit from their own group mean — noise that no treatment explains.

- Between: signal plus noise
- Within: noise only

CONSEQUENCE

F = between ÷ within

If H_o is true, both quantities estimate the same σ^2, and their ratio should be near 1. A ratio well above 1 means the group means are further apart than within-group noise can explain.

- F near 1: no evidence against H_o
- F large: reject H_o

Think about the other statistics like z-statistic, t-statistic. Does the concept still hold?

F-Statistic

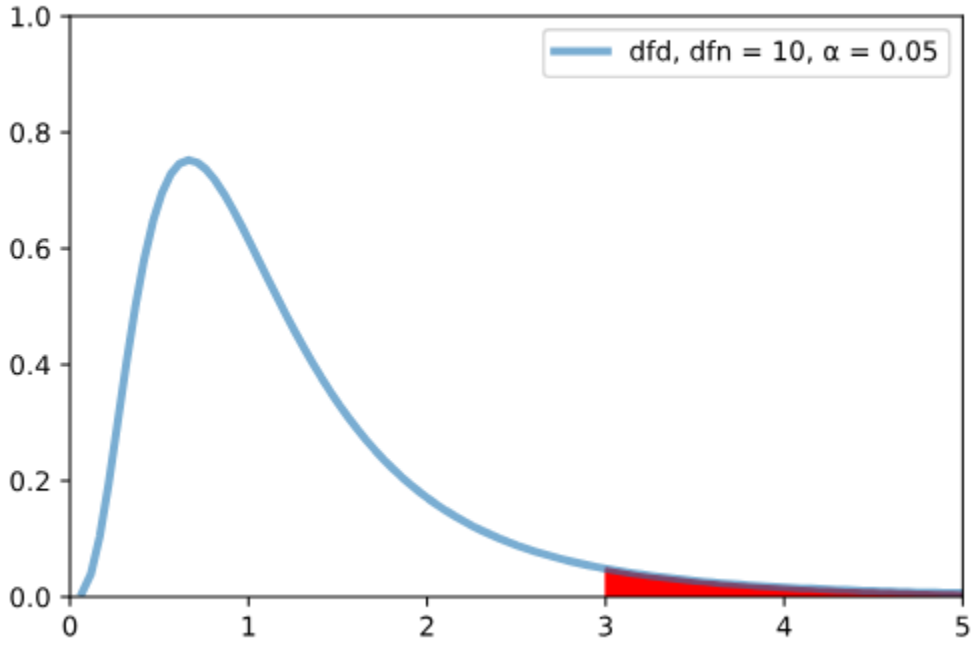
F-test

An F-test is a statistical test that compares variances. It is used to determine if the variances of two samples, or if the ratios of variances among multiple samples, are significantly different.

Null Hypothesis (H_0): $\sigma_1^2 = \sigma_2^2$ (The population variances are equal).

Alternative Hypothesis (H_1):

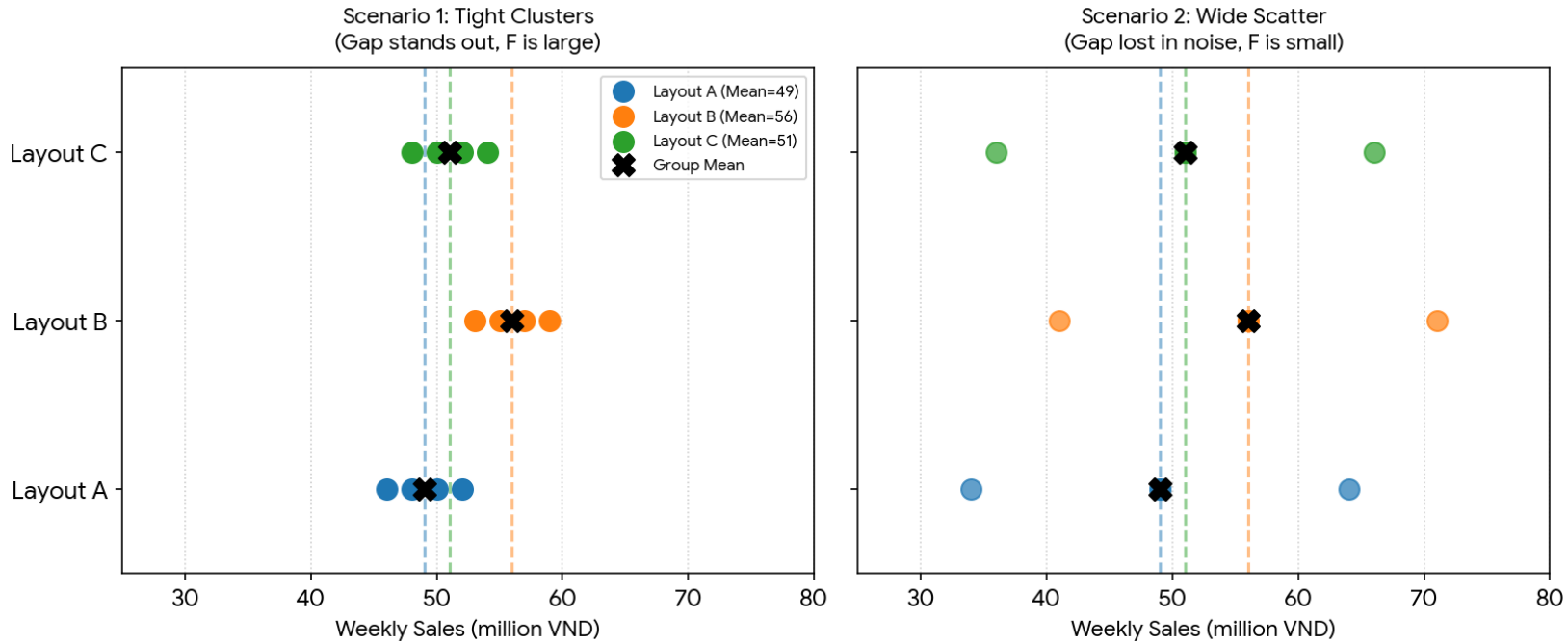
- $\sigma_1^2 \neq \sigma_2^2$ (Two-tailed)
- $\sigma_1^2 > \sigma_2^2$ (Right-tailed)
- $\sigma_1^2 < \sigma_2^2$ (Left-tailed)



Foundational mathematical engine behind **ANOVA (Analysis of Variance)** and **Multiple linear regression**

ANALYSIS OF VARIANCE

Same Means, Different Conclusions



WHY THE WITHIN-GROUP SPREAD MATTERS

The group means alone decide nothing

Two datasets with identical group means 49, 56 and 51. In the first, stores within each layout are tightly clustered; in the second, they are widely scattered.

- Tight clusters: between-group gap stands out, F large
- Wide scatter: the same gap is lost in noise, F small

The Decomposition of Variation

SSB**(Sum of Squares Between - Tổng bình phương giữa các nhóm):**

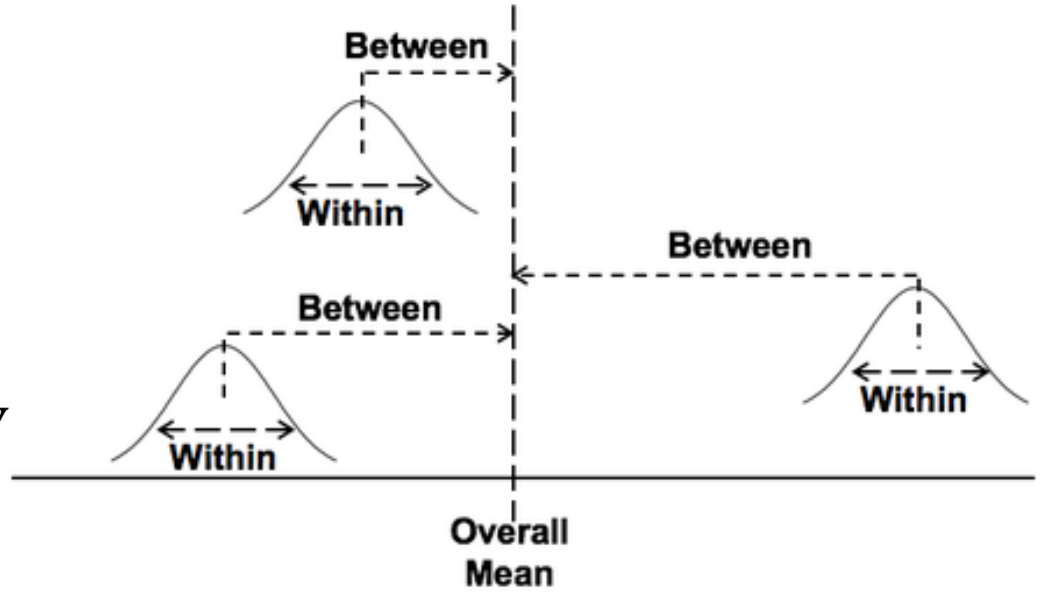
$$SSB = \sum n_i \, (\bar{x}_i - \bar{x})^2 \quad \text{(df: } k - 1\text{)}$$

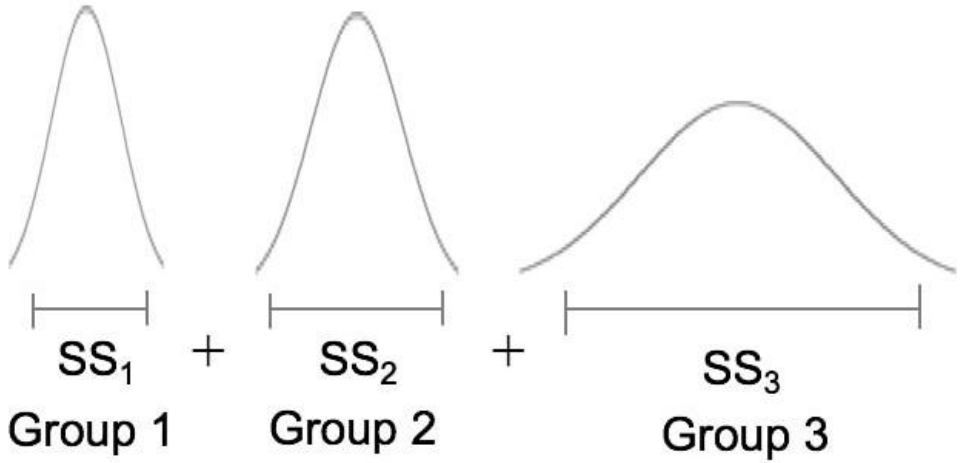
SSW**(Sum of Squares Within - Tổng bình phương nội bộ nhóm):**

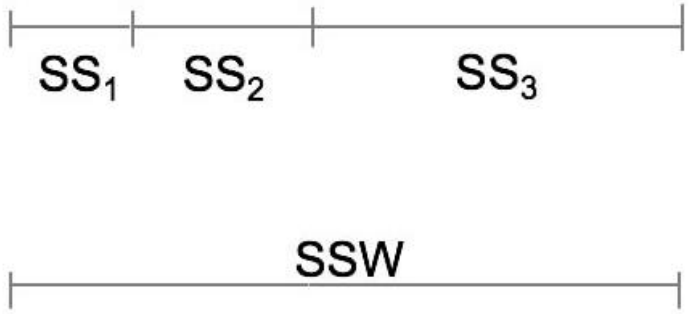
$$SSW = \sum (n_i - 1) \, s_i^2 \quad \text{(df: } N - k\text{)}$$

SST**(Sum of Squares Total - Tổng bình phương):** $SST = SSB + SSW$

Eta-squared: $\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} = \frac{SSB}{SST}$







The Decomposition of Variation

MSB**(Mean Squares Between Groups- Trung bình bình phương giữa các nhóm):**

$$MSB = \frac{SSB}{k-1}$$

MSW**(Mean Squares Within Groups- Trung bình bình phương nội bộ nhóm):**

$$MSW = \frac{SSW}{N-k}$$

$$\boldsymbol{F = \frac{Explained\ Variance}{Unexplained\ Variance}} = \frac{MSB}{MSW}$$

Under H_o and normal populations with equal σ, F has an F distribution with (k − 1, N − k) degrees of freedom.

ANALYSIS OF VARIANCE

Computing the Sums of Squares

STEP 1

Grand mean

$\bar{x} = (49 + 56 + 51)/3 = 52$, since all groups have n = 4

STEP 2

SSB

$4(49 - 52)^2 + 4(56 - 52)^2 + 4(51 - 52)^2 = 4(9 + 16 + 1) = 104$

STEP 3

SSW

Deviations in A: −1, 3, 1, −3, so 20. Same for B and C. SSW = 60

STEP 4

SST

104 + 60 = 164, the total squared deviation of all 12 stores from 52

STEP 5

Mean squares

MSB = 104/2 = 52. MSW = 60/9 = 6.67

ANALYSIS OF VARIANCE

The ANOVA Table

The standard layout in every statistical package, including Excel's ANOVA: Single Factor

Source	Sum of Squares	Degree of Freedom	Mean square	F
Between layouts	104	2	MSB = 52	F = 52 / 6.67 = 7.8
Within layouts (error)	60	9	MSW = 6.67	
Total	164	11		

F = 7.8 with (2, 9) df: is it far enough above 1?

The ANOVA Test, Step by Step

STEP 1

Hypotheses

$H_0 : \mu_A = \mu_B = \mu_C$ versus H_1 : not all three are equal

STEP 2

Test statistic

F = MSB/MSW, with (k − 1, N − k) = (2, 9) df

STEP 3

Rejection region

Right tail only: R : F ≥ $F_{0.05}(2, 9)$ = 4.26

STEP 4

Compute

F = 52/6.67 = 7.8

STEP 5

Conclude

7.8 ≥ 4.26: H_0 is rejected. p = 0.011. The layouts are not all alike

The F-test of ANOVA is always one-tailed: only a large F is evidence against H_0.

ANOVA Is the Pooled t-Test for k Groups

MSW = SSW/(N − k) is the pooled variance s_p^2, extended to k groups.

With k = 2, the ANOVA F equals the square of the pooled t:

ANOVA therefore carries the same assumptions as the pooled t-test: independent random samples, approximately normal populations, and a common σ. Here the three sample variances are equal, so the assumption is comfortable. When group variances differ strongly, the same warning as in the branch example applies, and a version of ANOVA that does not pool — Welch's ANOVA — is used instead.

Which Layouts Differ? Bonferroni Comparisons

Three pairs, so each is tested at $\alpha/3 = 0.0167$ · $SE = \sqrt{(MSW(1/4 + 1/4))} = 1.83$, $df = 9$

Pair	Difference · t	p-value	At $\alpha/3 = 0.0167$
B − A	7 · 3.83	0.004	different
B − C	5 · 2.74	0.023	not different — though $p < 0.05$
C − A	2 · 1.10	0.30	not different
Only B versus A survives the correction			

Bonferroni: with m comparisons, test each at α/m. The overall false-alarm rate then stays at or below α.

ANOVA and Bonferroni: Two Questions, Used in Sequence

ANOVA

Is there any difference at all?

One test of all k means together, with the false-alarm rate held exactly at α. A rejection says the treatments are not all alike, without naming which.

- Run first, on all groups at once

BONFERRONI

Which pairs differ?

Pairwise tests with the threshold lowered to α/m. It names the differing pairs, at the price of missing real but small differences more often as m grows.

- Run after ANOVA rejects

Later we will find the F-test of a regression on industry dummies is this same one-way ANOVA.

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Floor area enters; the ANOVA block of a regression output; $F = t^2$

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MULTIPLE REGRESSION

Both kinds of variable together: partial coefficients, t versus F, adjusted R^2

Writing a Category as Numbers

D_B = 1 if the store uses layout B, 0 otherwise. D_C = 1 if it uses layout C, 0 otherwise. A store with D_B = D_C = 0 uses layout A — the reference group.

A regression needs numbers, and a category is not a number. A dummy variable is a yes-or-no question coded as 1 or 0. Three layouts need only two questions: a store that answers no to B and no to C must be an A store.

ANOVA AS A REGRESSION

The Twelve Stores, Coded

Two new columns; the sales figures are unchanged from Session 10

Layout	D_B	D_C	Weekly sales of the four stores
A	0	0	48 · 52 · 50 · 46
B	1	0	55 · 57 · 53 · 59
C	0	1	50 · 54 · 52 · 48
Model: Sales = $b_0 + b_1 \cdot D_B + b_2 \cdot D_C + e$			

Reading the Coefficients

Coefficient	Estimate	Equals	Reading
b_0 (intercept)	49	mean of A	predicted sales of a layout-A store
b_1 (D_B)	7	56 − 49	B stores sell 7 more than A stores, on average
b_2 (D_C)	2	51 − 49	C stores sell 2 more than A stores, on average
Every dummy coefficient is a gap to the reference group			

ANOVA AS A REGRESSION

Two Outputs, One F

The same twelve stores, run through two different Excel tools

	ANOVA: Single Factor (Lecture 10)	Regression on D_B and D_C
Explained SS · df	Between groups: 104 · 2	Regression: 104 · 2
Unexplained SS · df	Within groups: 60 · 9	Residual: 60 · 9
F · p-value	7.8 · P-value 0.011	7.8 · Significance F 0.011
Share explained	$\eta^2 = 104/164 = 0.634$	R Square = 0.634

One-way ANOVA is a regression on dummy variables.

Why the Two Must Agree

MECHANISM

Knowing only the group, the best guess is its mean

OLS Regression minimizes the sum of squared residuals. A model with only dummies can output one value per group, and the value that minimizes squared deviations within a group is its mean.

- Fitted value = own group mean ($\hat{y}_i = \bar{y}_i$)
- Residual = distance from own group mean ($e_i = y_i - \hat{y}_i$)

CONSEQUENCE

Every ANOVA quantity reappears

Residual SS is SSW, regression SS is SSB, R^2 is η^2, and F is the same ratio.

The assumptions are similar:

- independent stores
- roughly normal errors
- one common σ for all groups

The Dummy Variable Trap

With k categories, use k − 1 dummies. Adding a third dummy D_A makes D_A + D_B + D_C = 1 for every store — an exact copy of the intercept column.

The model then has infinitely many equally good answers: any amount can be moved out of the intercept and added to all three dummies without changing a single prediction. The software cannot choose, and may not say so clearly.

k categories → k − 1 dummies. The category left out is the reference group.

Changing the Reference Group

REFERENCE = A

$b_0 = 49$, D_B = +7, D_C = +2. The coefficients answer: how do B and C compare with A?

REFERENCE = C

$b_0 = 51$, D_A = −2, D_B = +5. The coefficients answer: how do A and B compare with C?

B versus C now has its own standard error and t.

WHAT DOES NOT CHANGE

Fitted values, residuals, SSR = 104, $R^2 = 0.634$, F = 7.8. The model is the same; only the question each coefficient answers is different.

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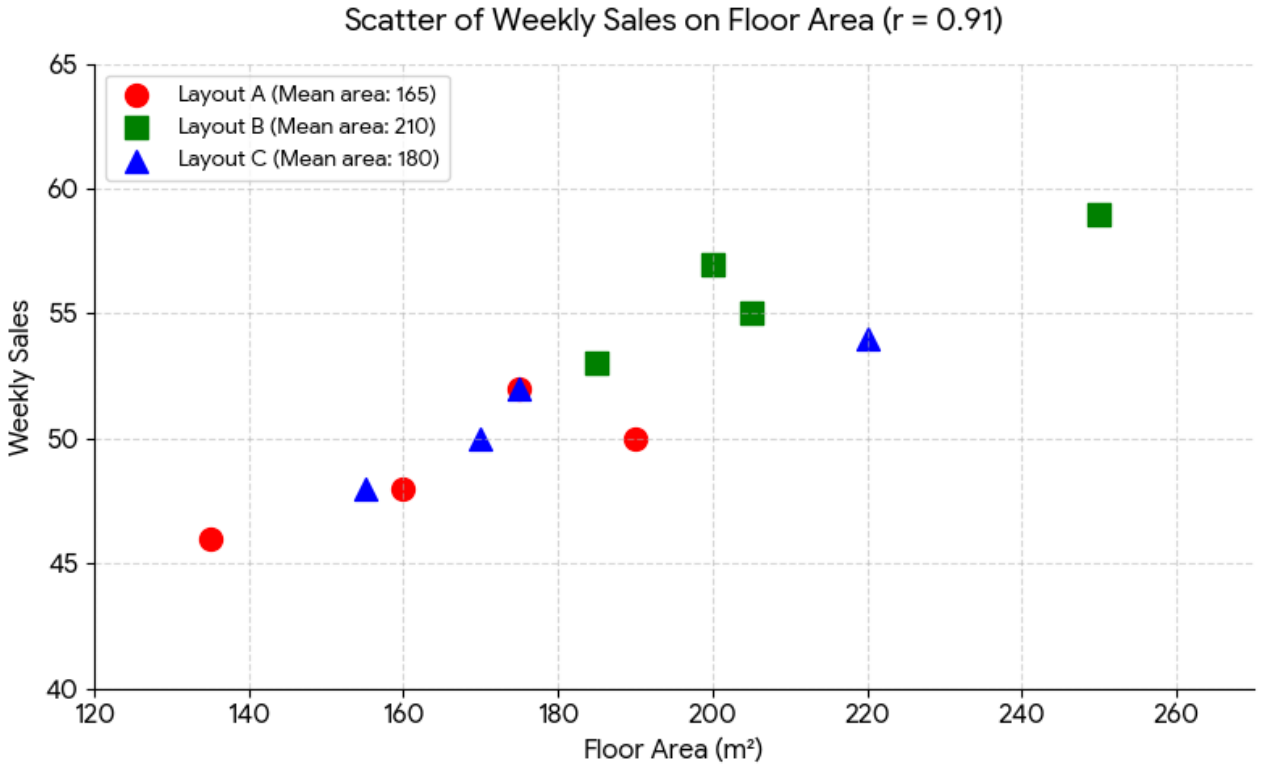
Store Size Enters the Picture

A SECOND VARIABLE

Floor area of the same twelve stores, in m²

Listed in the same order as the sales figures. Larger stores sell more: the correlation between weekly sales and floor area is r = 0.91.

- A: 160 · 175 · 190 · 135 (mean 165)
- B: 205 · 200 · 185 · 250 (mean 210)
- C: 170 · 220 · 175 · 155 (mean 180)



The B stores are also the largest. What does that do to the previous gaps between groups?

ONE CONTINUOUS PREDICTOR

The ANOVA Block of a Regression Output

Weekly sales on floor area · the top block of Excel's Regression output

	df	SS	MS
Regression	1	136.5	136.5
Residual	10	27.5	2.75
Total	11	164.0	
$F = 136.5 / 2.75 = 49.6$ · Significance $F < 0.001$ · $R^2 = 136.5 / 164 = 0.83$			

With One Predictor, $F = t^2$

Slope of floor area: b = 0.114, SE = 0.016, t = 7.04.

And $7.04^2 = 49.6 = F$

With a single X, ‘does the model explain anything?’ and ‘is the slope zero?’ are the same question, so the two tests must agree F equal the squared pooled t for two groups in Lecture 10. From two predictors onward, the two questions separate.

Two Kinds of X, One Machine

CATEGORICAL X

Fitted value = the group mean

The model predicts one value per group. Explained variation is the spread of the three group means around 52.

- SSR = SSB = 104 of 164

CONTINUOUS X

Fitted value = a point on the line

The model predicts a value for every floor area. Explained variation is the spread of the line's predictions around 52.

- SSR = 136.5 of 164

Both split SST into SSR + SSE, and both test the split with F = MSR / MSE.

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MULTIPLE REGRESSION

Three Models, One Coefficient

Weekly sales of the twelve stores, million VND

Model	Layout B vs A	Floor area (per m^2)	R^2 · adj. R^2
(1) Layout only	7.00	—	0.634 · 0.553
(2) Floor area only	—	0.114	0.832 · 0.816
(3) Layout + floor area	3.17	0.085	0.914 · 0.882
Holding floor area fixed cut the layout B premium from 7.0 to 3.2			

Where did the gap go?

7.00 = 3.17 + 0.085 × 45

The raw gap between B and A = the gap between stores of equal size + the area coefficient × how much larger the B stores are.

B stores average 210 m², A stores 165 m²: 45 m² more. At 0.085 million per m², size alone accounts for 3.83 of the 7.00. Model (1) credited all of it to the layout. This is omitted variable bias, computed exactly.

MULTIPLE REGRESSION

Ceteris Paribus, Made Concrete

MECHANISM

A partial coefficient compares like with like

In model (3), the B coefficient is the gap between B and A stores of the same floor area. The area coefficient is the gap per m^2 between stores using the same layout.

- Partial regression coefficient

CONSEQUENCE

The interpretation sentence changes

‘Among stores of the same floor area, layout B stores sell on average 3.2 million VND more per week than layout A stores.’ The clause on floor area is the whole content.

- Still no causal verb — only correlation

Randomized, Yet Unbalanced

Lecture 10 assigned the layouts at random. Random assignment balances store size on average — not in every sample of four stores.

Here the four B stores happened to be the largest. A control variable repairs imbalance in what was measured. Randomisation at scale protects against what was not measured. The two complement each other; neither replaces the other.

MULTIPLE REGRESSION

Model (3): The Coefficient Block

Sales = b_0 + b_1·D_B + b_2·D_C + b_3·Area + e · reference group: layout A

Variable	Coefficient	SE · t	p-value
Intercept	34.97	2.83 · 12.4	< 0.001
D_B (layout B)	3.17	1.20 · 2.64	0.030
D_C (layout C)	0.72	0.97 · 0.75	0.477
Floor area (m²)	0.085	0.017 · 5.11	0.001
$F(3, 8) = 28.4$, $p < 0.001$ · $R^2 = 0.914$ · adj. $R^2 = 0.882$ · n = 12			

Three Questions, Three Tests

Question	Test	Result
Does the model explain anything at all?	overall F(3, 8)	28.4 · p < 0.001
Do B stores differ from A stores of the same size?	t for D_B	2.64 · p = 0.030
Does layout matter at all, once size is held fixed?	partial F(2, 8) for D_B and D_C together	3.82 · p = 0.069
The partial F is Session 10's ANOVA, run after floor area has taken its share		

How can B differ from A, while layout as a whole is not significant at 5%?

Adjusted R^2: A Price for Each Variable

R^2 never falls when a variable is added — even a column of random numbers. Adjusted $R^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$ charges a price for each of the k predictors.

Model (3): $1 - (1 - 0.914) \times 11/8 = 0.882$. Adjusted R^2 compares models of different size on the same data. It is not a test, and a high value does not make a model right: model (1) had $R^2 = 0.63$ and a biased layout coefficient.

MULTIPLE REGRESSION

One Machine, Four Uses

The same SST = 164, split four ways

STEP 1

One categorical X: ANOVA

Group means as predictions. SSB = 104; F = 7.8

STEP 2

The same X as k − 1 dummies

Regression gives SSR = 104 and the same F; $R^2 = \eta^2$

STEP 3

One continuous X

A line as predictions. SSR = 136.5; $F = t^2 = 49.6$

STEP 4

Several X of both kinds

SSR = 149.9; partial coefficients, t per variable, F per group

PRACTICE ON VN-LISTED

Three Things to Take Away

1. ONE DECOMPOSITION

Every model splits the same SST into explained and unexplained. ANOVA and regression differ only in what does the explaining.

2. DUMMIES ARE GAPS

A dummy coefficient is a gap to the reference group, and one-way ANOVA is a regression on k − 1 dummies.

3. CONTROLS CHANGE THE QUESTION

A control changes what a coefficient compares: B versus A became B versus A at equal size